The Sector Anatomy of Commodity Risk Premia*

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Abstract

We sort commodity futures on the sector-wide and sector-adjusted components of their characteristics, defined as the fitted values and residuals of monthly cross-sectional regressions on sector indicators. The sector-wide component accounts for most of the variance of characteristic-managed returns, but the pricing content that standard commodity factor models leave unexplained is sector-adjusted: sector-adjusted portfolios earn alphas against these models and sector-wide portfolios do not. Latent factors extracted from the sector-adjusted portfolios price the original portfolios better than those from the sector-wide portfolios. A characteristic-managed return can therefore comove mostly with sector returns while its unexplained premium is earned within sectors.

Keywords: Asset Pricing, Commodity Markets, Risk Premia, Latent Factors, Factor Models, Characteristic-Managed Portfolios.

JEL codes: G11, G12, G13

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1 Introduction

In the hedging-pressure view of commodity futures pricing (Keynes, 1930; Hirshleifer, 1988; De Roon et al., 2000), premia compensate speculators for absorbing the price risk that hedgers with physical exposures transfer to them. Because commodity futures markets are only partially integrated with broader capital markets (Bessembinder, 1992; Hong and Yogo, 2012; Acharya et al., 2013), this compensation depends on the traders active in each market, such as producers, merchants, swap dealers, and financial speculators, and need not be tied to an aggregate market portfolio.

Characteristics such as momentum, the futures basis, and basis-momentum describe the cross-section of these premia (Erb and Harvey, 2006; Szymanowska et al., 2014; Boons and Porras Prado, 2019). Commodities, however, trade in a small number of sectors. A characteristic measured commodity by commodity therefore reflects conditions common to the sector, such as aggregate energy demand or a sector-wide inventory cycle, as well as conditions specific to the commodity, such as local storage constraints or idiosyncratic supply shocks. Which of the two carries the premium, and whether they are priced through the same mechanism, is the question of this paper.

A similar question arises in equity markets. Moskowitz and Grinblatt (1999) find that industry momentum accounts for a large share of individual stock momentum, and Asness et al. (2000) find that industry-relative value signals predict returns more reliably than their unadjusted counterparts. Whether these patterns carry over to commodity futures is not obvious, since commodity sectors are few and are defined by the physical characteristics of the underlying commodities. If commodity momentum were primarily a sector phenomenon, explanations based on the gradual diffusion of information across related assets (Hong and Stein, 1999; Hong et al., 2007) would gain support; if it survives sector adjustment, explanations that operate at the level of the individual asset become more plausible.

We first document that sector membership matters for both comovement and expected returns. In our sample of 46 commodity futures from 1987 to 2024, the equal-weighted portfolio of a commodity's own sector explains a median 45% of the variance of its monthly return, and the other sector portfolios add almost nothing, so the remaining variation is commodity-specific. In panel regressions of returns on five widely studied characteristics, sector fixed effects add little explanatory power and are not significant, so sector membership on its own does not predict returns. Interactions between the characteristics and the sector indicators account for most of the explained variation, so a characteristic's predictive content depends on the commodity's sector. Much of this sector dependence runs through the sector-level average of each signal in each month.

This evidence motivates the decomposition we use throughout the paper. Adapting Langlois (2023), we decompose each characteristic through a cross-sectional regression on sector indicators. The fitted value, which is the average signal of the commodity's sector, is the sector-wide component; the residual is the sector-adjusted component. Sorting on each component in the full cross-section yields two families of characteristic-managed portfolios that together retain both the sector-level and the commodity-level variation of the signal. Neither of the two common ways of handling sector effects does so: an equal-weighted sector portfolio ignores the characteristic, and a within-sector sort discards the cross-sector dispersion of the signal and so cannot identify wider sector-level effects.

At the portfolio level, the two components divide the variance and the pricing content of characteristic-managed returns differently. The sector-wide component carries a median of 32% of the cross-sectional variance of the signals, yet it accounts for a median 61% of the variance of the unadjusted portfolio return under linear weights. Tercile sorts give the same ordering: the unadjusted return has a median spanning R^2 of 44% on the sector-wide component against 29% on the sector-adjusted one. The asymmetry is specific to economic sectors. Over 1,000 random partitions of the commodities into groups of the same sizes, the sector-wide share of return variance falls to a median of 0.11, about the same as the

between-group share of signal variance.

Spanning tests show that neither component alone prices the unadjusted returns, while the two together leave only 4 of the 46 characteristic-managed portfolios with an alpha significant at the 5% level. To examine whether the two components carry distinct priced risk, we estimate the pricing kernel with the risk-premium PCA of Lettau and Pelger (2020a). This method extracts the factors that explain both comovement and cross-sectional differences in expected returns, without imposing observable factors on a partially segmented market. The question is whether the sector-wide and sector-adjusted channels are priced differently. Factors extracted from the sector-adjusted portfolios price the unadjusted portfolios with a cross-sectional R^2 of 75% at seven factors, against 60% for factors extracted from the sector-wide portfolios, with comparable pricing errors. The ranking holds for every number of factors, and when the closely related characteristics are pruned to one per group.

We also test both channels against observable commodity factor models. Sector-wide portfolios are absorbed by standard three- and five-factor commodity benchmarks, with or without the commodity market factor, whereas sector-adjusted portfolios reject all four, as well as benchmarks built from the sector-adjusted characteristics themselves. The pricing content that the decomposition adds to the existing factor literature is therefore concentrated in the within-sector channel, which no small set of characteristic-sorted factors spans, whatever their construction.

The two channels also differ over the business cycle. The median Sharpe ratio of sector-wide high-minus-low portfolios rises in NBER recessions, and that of sector-adjusted portfolios falls. In CFTC positioning data, managed money offsets producer hedging demand at both the sector and the commodity level, consistent with risk transfer from hedgers to speculators at both levels.

The decomposition also follows from an equilibrium model of risk transfer (Appendix A). When hedgers with net physical exposure trade with speculators, the sector structure of

payoff risk separates premia into exactly these two components, and sector-wide exposure is priced more per unit (Proposition 1). The model has two kinds of implications. The first concerns the pattern of risk transfer, and the data are consistent with it: speculators take the other side of hedgers' positions at both levels, the slopes of premia on positions have the predicted signs and ordering, although they are not significant, and the within-sector dispersion of hedging positions compresses in recessions. The second concerns the source of the premia, which in the model is hedgers' net exposure. The portfolio built from hedgers' sector-adjusted positions does not price the sector-adjusted portfolios (GRS p-values of at most 0.004), so hedging exposure, as measured in the positions data, does not account for the sector-adjusted premium. What drives that premium remains open.

Momentum deserves special attention because its premium divides evenly between the two channels. Its sector-wide and sector-adjusted components earn 5.21% and 5.22% a year under linear weights, even though most of the variance of momentum returns is sector-wide. Commodity momentum is thus not primarily a sector phenomenon, and the sector-adjusted half calls for an explanation at the level of the individual commodity. We test four candidates: gradual diffusion of information across related commodities, slow-moving capital (Duffie, 2010), informational frictions, and persistent hedging imbalances. None of them accounts for the sector-adjusted premium: sector-adjusted returns do not predict subsequent sector returns, as gradual diffusion requires, the interactions with the costs of arbitrage and with informational frictions are not significant, and controlling for hedging imbalances changes the momentum coefficient little.

Our results contribute to two literatures. In the asset pricing literature on commodities (Erb and Harvey, 2006; Szymanowska et al., 2014; Boons and Porras Prado, 2019; Sakkas and Tessaromatis, 2020), we show that standard characteristic sorts combine a sector-wide component that accounts for most of their variance with a nearly uncorrelated sector-adjusted component that carries the premium these factor models leave unexplained. Prior work on sector structure in commodity markets studies the comovement of returns and of futures

curves (Daskalaki et al., 2014; Karstanje et al., 2017; Christoffersen et al., 2019); we study how that structure divides characteristic premia and their pricing. In the equity literature on industry effects in characteristics (Moskowitz and Grinblatt, 1999; Asness et al., 2000; Cohen and Lou, 2012), we provide evidence from a differently structured asset class, in which the within-group and across-group components of characteristic premia are priced differently, but the momentum premium is concentrated in neither.

2 Data and the Sector Decomposition

We describe the futures data and the characteristics, document how sector membership relates to the comovement and predictability of commodity returns, and then define the decomposition that this evidence motivates.

2.1 Data

Our dataset consists of 46 US dollar-denominated commodity futures in five sectors: Energy, Grains & Oilseeds, Livestock, Metals, and Softs. For each contract, we collect end-of-day settlement prices, open interest, and trading volume from Bloomberg. Table IA.1 of the Internet Appendix details the contracts' sectors, trading exchanges, Bloomberg tickers, first observation years, delivery months, and CFTC codes.[1]

The return on a fully collateralized futures position is calculated as:

$$r_{i,t+1} = \frac{F_{i,t+1}^{(T)} - F_{i,t}^{(T)}}{F_{i,t}^{(T)}} + r_t^f, \tag{1}$$

where $F_{i,t+1}^{(T)}$ and $F_{i,t}^{(T)}$ are the futures prices of commodity i at months $t+1$ and t for the nearby contract expiring at time T (Gorton et al., 2013; Yang, 2013; Koijen et al., 2018).

[1]Our panel of futures contracts is comparable to or more extensive than those used in previous studies, including Hong and Yogo (2012), Gorton et al. (2013), Daskalaki et al. (2014), Szymanowska et al. (2014), Bakshi et al. (2019), Boons and Porras Prado (2019), and Sakkas and Tessaromatis (2020).

The term r_t^f represents the interest earned on the collateral.

Following Szymanowska et al. (2014) and Bakshi et al. (2019), we hold a long position in the nearest-to-maturity contract and roll it into the second nearby contract by closing the first position and opening the second on the same day. When the last trading day of a contract falls in the month before its contract month, we roll on the last trading day. When it falls in or after the contract month, we roll on the first day of the contract month, so that the position never carries a physical delivery obligation (Hong and Yogo, 2012). The excess return for commodity i is then defined as:

$$xr_{i,t+1} = r_{i,t+1} - r_t^f. \tag{2}$$

Figure IA.1 in Section IA.A of the Internet Appendix illustrates our dataset coverage from August 1959 to June 2024, with Panel A showing the available commodity futures and Panel B the available characteristic-managed portfolios.

Our main empirical analysis covers March 1987 to June 2024, a period with enough contracts in every month for tercile sorts (Panel A) and 46 characteristics available throughout (Panel B). The start date of the main sample fixes the number of characteristics: more characteristics, and hence more test assets, can raise statistical power but shorten the time series, whereas a longer time series improves estimation precision but excludes characteristics unavailable in earlier periods, so the choice of sample period trades breadth against length. Section 6 examines a longer sample from August 1969 to June 2024 with 40 characteristics and a shorter sample from January 1996 to June 2024 with 52.

Section IA.A of the Internet Appendix presents the descriptive statistics for the monthly futures excess returns. Returns (Table IA.2) and correlations (Figure IA.2) differ systematically within and across sectors. Average correlations are higher within sectors than across sectors in four of the five sectors and similar in Softs (0.10 within and 0.11 across), consistent with sector exposures contributing to the comovement of commodity futures returns.

2.2 Characteristics

We begin by constructing 58 characteristics, organized into four categories by their economic content. A characteristic enters a sample only if at least 10 commodities carry it in every month of that sample, the minimum a tercile sort accepts; 46 characteristics meet this condition in the main sample. *Fundamental* characteristics capture systematic risk exposures (market beta, USD beta, inflation beta), the term structure of futures prices (basis, basis-momentum, relative basis), and the positions of hedgers and speculators (hedging pressure, speculative pressure). *Trading frictions* characteristics measure liquidity and trading activity (volume, turnover, open interest growth, net trading). *Past performance* characteristics include momentum signals at horizons ranging from one month to five years, residual momentum, and reversal and seasonality measures. *Downside risk and volatility* characteristics capture higher moments (skewness, kurtosis), tail risk, and various beta specifications (downside beta, betting-against-beta).

Our characteristic set incorporates established measures from the commodity literature alongside signals adapted from equity studies (Freyberger et al., 2020). Table 1 lists all characteristics with sources; detailed construction procedures appear in Section IA.B of the Internet Appendix.

2.3 Sector Structure and the Characteristic Decomposition

We begin by measuring how much of the variation in commodity returns is sector-level. For each commodity, we regress monthly excess returns on the equal-weighted portfolio of its own sector, and then jointly on all five sector portfolios. Section IA.C.1 of the Internet Appendix shows the results. The own-sector portfolio explains a median 45.4% of return variance, and adding the other four sectors raises the median only to 45.9%. Sector exposure is therefore a large but incomplete source of comovement, and the remaining variation appears commodity-specific rather than spanned by other sectors, consistent with the weak factor structure in

#	Definition	Abbreviation	Source
	Fundamental		
(1)	market beta	mkt_beta	Fama and MacBeth (1973)
(2)	USD beta	usd_beta	Szymanowska et al. (2014)
(3)	inflation beta	infl_beta	Szymanowska et al. (2014)
(4)	value	value	Asness et al. (2013)
(5)	basis	basis	Erb and Harvey (2006)
(6)	basis momentum	basis_mom	Boons and Porras Prado (2019)
(7)	relative basis	rel_basis	Gu et al. (2019)
(8)	hedging pressure	hedge_press	Basu and Miffre (2013)
(9)	speculative pressure	spec_press	Fan et al. (2020)
	Trading frictions		
(10)	abnormal turnover	abn_turn	Connolly and Stivers (2003)
(11)	trading volume	volume	Chordia et al. (2001)
(12)	turnover	turnover	Amihud and Mendelson (1986)
(13)	de-trended turnover	de_trend_turn	Chen et al. (2001)
(14)	volume trend	vol_trend	Gallant et al. (1992)
(15)	volume volatility	vol_vola	Chordia et al. (2001)
(16)	coeff. of variation of turnover	coefvar_turn	Chordia et al. (2001)
(17)	coeff. of variation of dollar volume	coefvar_vol	Chordia et al. (2001)
(18)	growth rate open interest	growth_oi	Hong and Yogo (2012)
(19)	CAPM alpha	capm_alpha	Fama and MacBeth (1973)
(20)	illiquidity	illiq	Amihud (2002)
(21)	zero trading days	zero_days	Liu (2006)
(22)	net trading	net_trading	Kang et al. (2020)
(23)	idiosyncratic volatility	idio_vol	Ali et al. (2003)
(24)	idiosyncratic skewness	idio_skew	Yao et al. (2019)
(25)	information discreteness	info_discret	Huang et al. (2022)
(26)	momentum high volume	mom_high_vol	Lee and Swaminathan (2000)

#	Definition	Abbreviation	Source
	Past performance		
(27)	short-term reversal	mom_2_1	Cheng et al. (2017)
(28)	momentum 3 mths.	mom_3	Miffre and Rallis (2007)
(29)	momentum 6 mths.	mom_6	Miffre and Rallis (2007)
(30)	momentum 9 mths.	mom_9	Miffre and Rallis (2007)
(31)	momentum 12 mths.	mom_12	Miffre and Rallis (2007)
(32)	momentum 12 mths. to 7 mths.	mom_12_7	Novy-Marx (2012)
(33)	momentum 24 mths.	mom_24	Miffre and Rallis (2007)
(34)	momentum 36 mths.	mom_36	Miffre and Rallis (2007)
(35)	momentum 60 mths.	mom_60	Miffre and Rallis (2007)
(36)	long term reversal	lt_reversal	De Bondt and Thaler (1985)
(37)	seasonality	season	Heston and Sadka (2008)
(38)	52-week high	weeks_52w	George and Hwang (2004)
(39)	change 6m momentum	chg_6m_mom	Gettleman and Marks (2006)
(40)	residual momentum 6 mths.	resid_mom_6m	Blitz et al. (2011)
(41)	residual momentum 12 mths.	resid_mom_12m	Blitz et al. (2011)
	Downside risk and volatility		
(42)	skewness	skew	Amaya et al. (2015)
(43)	kurtosis	kurtosis	Amaya et al. (2015)
(44)	volatility	volatility	Schneider et al. (2020)
(45)	co-skewness	co_skew	Harvey and Siddique (2000)
(46)	co-kurtosis	co_kurt	Dittmar (2002)
(47)	coeff. of variation	coefvar	Szymanowska et al. (2014)
(48)	semi deviation	semi_dev	Post et al. (2006)
(49)	betting against beta	bab	Frazzini and Pedersen (2014)
(50)	betting against correlation	bac	Asness et al. (2020)
(51)	dimson beta	dimson	Dimson (1979)
(52)	downside beta	down_beta	Ang et al. (2006)
(53)	systematic tail risk	syst_tail_risk	Van Oordt and Zhou (2016)
(54)	tail risk	tail_risk	Kelly and Jiang (2014)
(55)	Cornish-Fisher VaR	var_cf	Füss et al. (2010)
(56)	VIX beta	vix_beta	Ang et al. (2009)
(57)	market correlation	mkt_correl	Pollet and Wilson (2010)
(58)	delay	delay	Lin et al. (2014)

Table 1: **Commodity characteristics by category.** This table lists the 58 characteristics used to construct test assets and risk factors. Characteristics are organized into four categories: Fundamental (market exposures, basis measures, positioning data); Trading frictions (volume, turnover, liquidity measures); Past performance (momentum at various horizons, residual momentum, reversal, seasonality); Downside risk and volatility (higher moments, tail risk, beta variants). For each characteristic, we report a brief definition, the abbreviation used throughout the paper, and the primary reference for its construction. Section IA.B of the Internet Appendix details the construction procedures.

commodity futures returns documented by Christoffersen et al. (2019).

To test whether sector membership also matters for expected returns, we regress commodity excess returns on five widely studied characteristics (Sakkas and Tessaromatis, 2020), namely momentum, hedging pressure, basis, basis-momentum, and value, together with sector indicators and sector–characteristic interactions. Section IA.C.9 of the Internet Appendix reports the estimates. The sector fixed effects are not significant at the 5% level, whereas the interactions account for 63% of the explained variation and are jointly significant ($p = 0.03$). Replacing the sector indicators with sector $\times$ time fixed effects, which absorb the part of the interactions that varies only by sector and month, lowers the R^2 contributed by the interactions from 0.48 to 0.27 percentage points, and they are no longer significant at the 5% level ($p = 0.09$). A substantial part of the predictive content of the interactions therefore operates through time-varying sector-level averages of the signals.

This evidence motivates the decomposition we use throughout the paper. Adapting Langlois (2023), for characteristic j in month t, we estimate:

$$z_{i,t}^{(j)} = \sum_{s=1}^{N_s} S_{s,t}^{(j)} \mathbb{I}_{i \in s} + \upsilon_{i,t}^{(j)}, \quad i = 1, \ldots, N_t \tag{3}$$

where $S_{s,t}^{(j)}$ is the contribution of sector s to characteristic j, $\mathbb{I}_{i \in s}$ indicates sector membership, and $\upsilon_{i,t}^{(j)}$ is the residual.[2] The decomposition yields two components:

$$\widehat{S}_{i,t}^{(j)} = \sum_{s=1}^{N_s} \widehat{S}_{s,t}^{(j)} \mathbb{I}_{i \in s} = \bar{z}_{s(i),t}^{(j)} \qquad \text{(sector average)} \tag{4}$$

$$\widehat{\upsilon}_{i,t}^{(j)} = z_{i,t}^{(j)} - \bar{z}_{s(i),t}^{(j)} \qquad \text{(deviation from sector average)} \tag{5}$$

where $s(i)$ denotes the sector to which commodity i belongs.[3] We apply the decomposition

[2] This specification without intercept treats all sectors symmetrically. The OLS first-order condition $\partial / \partial S_{s,t}^{(j)} \sum_{i \in s} (z_{i,t}^{(j)} - S_{s,t}^{(j)})^2 = 0$ implies $\hat{S}_{s,t}^{(j)} = \bar{z}_{s,t}^{(j)}$, i.e., each coefficient equals the sector-s average of characteristic j.

[3] Our approach is related to industry-adjustment methods in equities: Moskowitz and Grinblatt (1999) de-

in Equation (3) to the characteristic on its own scale.

Separating the two components permits a direct attribution of a characteristic's premium to sector-level and commodity-level content, and it allows the two components to carry different prices of risk. In the equilibrium model of Appendix A, hedgers transfer payoff risk with a common sector component to speculators, and sector-wide hedging exposure is priced more per unit than sector-adjusted exposure (Proposition 1). Section 5 tests the model's implications for positions, risk premia, and the business cycle.

Following Kelly et al. (2019), the original characteristic and both components are standardized by ranking, demeaning, and scaling to $[-0.5, 0.5]$, denoted $\widetilde{z}_{i,t}^{(j)}$, $\widetilde{S}_{i,t}^{(j)}$, and $\widetilde{v}_{i,t}^{(j)}$. Standardizing after the decomposition leaves portfolio composition unchanged, since tercile membership is invariant to monotone transformations, but it makes signal magnitudes comparable across characteristics and over time.[4]

The construction differs from the two approaches commonly used to handle sector effects, namely equal-weighted sector portfolios and within-sector sorts. An equal-weighted sector portfolio holds every commodity in the sector with the same weight, whatever its signal, so it captures sector exposure but cannot test whether characteristics predict returns within or across sectors. A within-sector sort ranks commodities in each sector separately. It isolates commodity-specific effects but discards cross-sector information, so a commodity ranking first in a low-dispersion sector receives the same weight as one ranking first in a high-dispersion sector. Because the long and short positions offset within each sector, it cannot identify sector-level effects. We instead sort both components in the full cross-section, so the sector-wide sort takes positions in whole sectors, and the sector-adjusted sort tests whether commodities above their sector average outperform those below it. Table IA.8 in Section IA.C.10 of the Internet Appendix illustrates the three sets of weights with a stylized

compose stock *returns* into industry and intra-industry components, whereas Asness et al. (2000) decompose firm *characteristics* into within-industry and across-industry components. We adapt the characteristic-level decomposition to the commodity sector structure.

[4]Standardizing before the decomposition would change the components themselves, because ranking is nonlinear, so the sector average of the ranks differs from the rank of the sector average.

momentum example.

3 Sector-Wide and Sector-Adjusted Risk Premia

We compare the risk premia of the three constructions and examine which component accounts for the comovement of the unadjusted characteristic-managed returns and whether either component alone prices them.

3.1 Risk Premia Across Portfolio Constructions

Our main empirical analysis uses, for each of the three constructions, 138 tercile portfolios formed on the 46 characteristics described in Section 2.[5] Of the 138 unadjusted portfolios, 81 have average returns significant at the 5% level with Newey–West standard errors with $\lfloor 4(T/100)^{2/9} \rfloor = 5$ lags, the rule used throughout the paper. The sector-wide and sector-adjusted portfolios of the same characteristic are close to uncorrelated: the median absolute correlation between their HML returns is 0.11, and 78% of the correlations lie between -0.2 and 0.2.

Unadjusted test asset portfolios span the widest risk-return range, sector-wide portfolios an intermediate one, and sector-adjusted portfolios cluster near the origin (Figure IA.4 in Section IA.C.2 of the Internet Appendix). The compression in both risk and return is consistent with sector exposures accounting for part of the commonality across commodity returns.

Panel A of Table 2 reports the premia of the HML portfolios. Median annualized risk premia are 2.99% for unadjusted portfolios, 1.46% for sector-wide portfolios, and 0.59% for sector-adjusted portfolios, consistent with both channels contributing to expected returns.

[5] Most previous studies use smaller sets of test assets: Bakshi et al. (2019) use 12 portfolios; Sakkas and Tessaromatis (2020) use 27 tercile portfolios based on 9 characteristics; Szymanowska et al. (2014) use 24 portfolios sorted on six characteristics; Sakkas (2024) use 33 tercile portfolios based on 11 characteristics; and Boons and Porras Prado (2019) use 16 portfolios sorted on four characteristics or 21 individual commodity futures.

The significance of the premia is ordered differently from their size: the sector-adjusted mean return is significant at the 5% level for 12 of the 46 characteristics, as often as the unadjusted one, while the sector-wide mean is significant for 6 characteristics.

Figure IA.5 in Section IA.C.3 of the Internet Appendix shows that sector-wide portfolios exhibit the highest volatility (median 20.45% annualized), followed by the unadjusted (16.28%) and sector-adjusted (12.5%) portfolios. Removing the sector-wide component compresses both returns and risk, whereas removing the sector-adjusted component lowers returns but raises risk. Median appraisal ratios (alpha divided by the residual volatility from a regression on the equal-weighted average of the 138 tercile portfolios of the same construction) are nonetheless similar across constructions, at 0.07 for the unadjusted and sector-wide portfolios and 0.08 for the sector-adjusted portfolios.

Over the sample, unadjusted HML portfolios produce the largest cumulative returns for momentum, basis, and basis-momentum, whereas for hedging pressure, 52-week high, and seasonality, a decomposed component earns a larger cumulative return. The sector-adjusted component earns the larger return for hedging pressure, and the sector-wide component for the other two (Figures IA.6 and IA.7 in Section IA.C.3 of the Internet Appendix). The relative importance of the two channels therefore differs across characteristics and, as the cumulative returns show, over time.

3.2 Variance Decomposition of the Unadjusted Returns

We measure which component accounts for more of the comovement of the unadjusted return by the R^2 of univariate spanning regressions of unadjusted HML returns on each component separately. Figure 1 presents the results across all 46 characteristics. In Panel A, 83% of characteristics lie below the 45-degree line, so sector-wide returns explain more of the variation in unadjusted returns, with a median R^2 of 0.44 against 0.29 for sector-adjusted returns. The slope coefficients are larger for the sector-adjusted component (median $\beta_{\text{SA}} =$

	Unadjusted	Sector-wide	Sector-adjusted
Panel A: Tercile portfolios			
Mean return (% p.a.), median	2.99	1.46	0.59
Sharpe ratio, median	0.16	0.07	0.05
Mean significant at 5%	12 of 46	6 of 46	12 of 46
Panel B: Decomposition of the unadjusted return			
Variance share θ, median		0.61	0.39
Larger variance share		39 of 46	7 of 46
Correlation of the components, median		0.07	
Univariate R^2, linear weights, median		0.63	0.43
Univariate R^2, terciles, median		0.44	0.29
Panel C: Linear-weight portfolios			
Mean return (% p.a.), median	2.72	0.80	0.39
Sharpe ratio, median	0.17	0.06	0.04
Mean significant at 5%	12 of 46	4 of 46	15 of 46

Table 2: **Risk premia and the decomposition of the unadjusted return.** Panel A reports, for the tercile high-minus-low portfolios of each construction, the median across the 46 main-sample characteristics of the annualized mean return and Sharpe ratio, and the number of characteristics whose mean is significant at the 5% level with Newey–West standard errors. Panel B reports the median variance share θ defined in Section 3.2, the number of characteristics for which each component has the larger share, the median correlation between the two components, and the median univariate R^2 of the unadjusted return on each component under linear weights and under the tercile construction. Sample: March 1987 to June 2024, 448 months. Panel C reports the same statistics for the linear-weight portfolios of Equation (6). Under linear weights, the mean returns of the two components add up to that of the unadjusted portfolio for each characteristic, although the medians across characteristics need not; the tercile portfolios hold different commodities in the three constructions, so their mean returns do not add up. The component mean returns are scaled differently in the two panels; Sharpe ratios do not depend on the scale.

0.73 against $\beta_{\text{SW}} = 0.52$; Figure IA.8 of the Internet Appendix), which reflects the lower volatility of sector-adjusted portfolios.[6]

The division between the two components varies across characteristics. For the basis, the two components have roughly balanced explanatory power (R^2 of about 0.25 each). Price delay and betting-against-correlation lie toward the opposite end, with univariate R^2 of 0.55 and 0.64 on the sector-wide component against 0.03 and 0.14 on the sector-adjusted component. Unadjusted momentum returns comove mostly with the sector-wide component

[6]Because variance explained scales with the regressor's variance, the higher sector-wide R^2 is the relevant comparison. More formally, $R^2 = \beta^2 \operatorname{Var}(r^x)/\operatorname{Var}(r^{\text{unadj}})$.

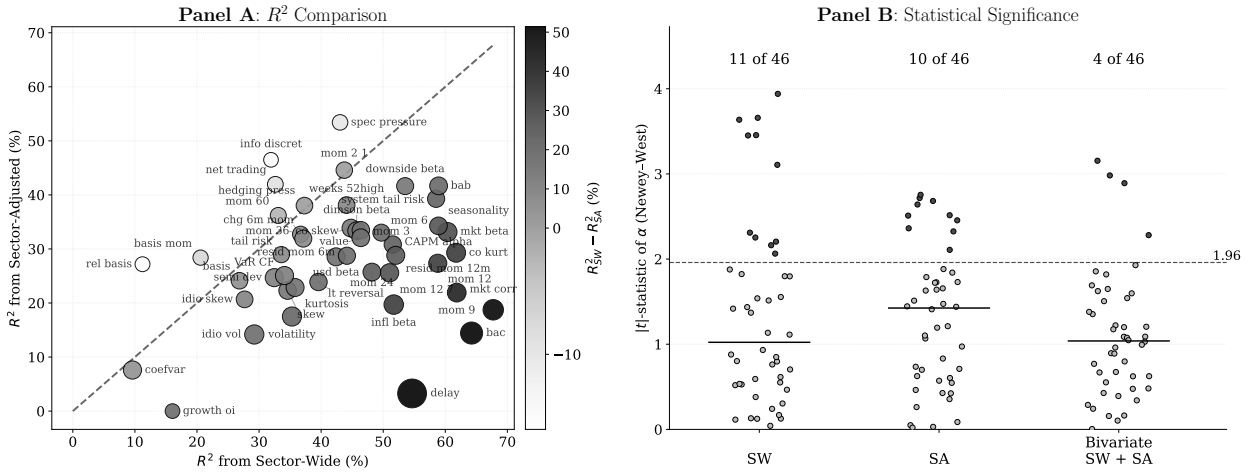


Figure 1: **Spanning regression results.** Panel A plots the R^2 from univariate regressions of unadjusted HML returns on sector-wide returns (horizontal axis) against sector-adjusted returns (vertical axis). Each point is one characteristic; its shade is the difference $R^2_{\text{SW}} - R^2_{\text{SA}}$, its size increases with the ratio of the slopes $\beta_{\text{SW}}/\beta_{\text{SA}}$, and the dashed line is the 45-degree line. Panel B plots, for each characteristic, the absolute Newey–West t-statistic of the intercept from the univariate regressions on each component and from the bivariate regression on both. The dashed line marks $|t| = 1.96$, darker points lie above it, the numbers report how many do, and the horizontal bars mark the medians. Sample: March 1987 to June 2024.

(R^2 of 0.59 against 0.27 in Table 3), although Section 5.4 shows that the momentum premium divides almost evenly between the two components.

Table 3 sets the sector-wide share of signal variance against the sector-wide share of return comovement for the five characteristics of the panel regressions of Section 2.3. The first is the fraction of the cross-sectional variance of the ranked signal due to its sector mean; the second is the univariate R^2 of the unadjusted HML return on the sector-wide return relative to the sum of the two univariate R^2 values. The sector mean accounts for between 16% (basis-momentum) and 33% (value) of signal variance at the median. The sector-wide return accounts for 68% of the comovement for momentum, 59% for value, 53% for basis, 50% for hedging pressure, and 42% for basis-momentum, so its share of return comovement exceeds its share of signal variance for all five characteristics. The larger role of the sector-wide component therefore does not follow from the dispersion of the signals.

	Signal variance		Return comovement		
Characteristic	SW share	[p25, p75]	R^2_{SW}	R^2_{SA}	SW share
Momentum (12 months)	0.31	[0.21, 0.43]	0.59	0.27	0.68
Hedging pressure	0.28	[0.17, 0.38]	0.37	0.38	0.50
Value	0.33	[0.19, 0.46]	0.46	0.32	0.59
Basis	0.18	[0.12, 0.30]	0.27	0.24	0.53
Basis-momentum	0.16	[0.09, 0.25]	0.21	0.28	0.42

Table 3: **Sector-wide characteristic signals vs. portfolio returns.** The "signal variance" columns report the median share of the monthly cross-sectional variance of each characteristic's ranked signal $\tilde{z}_{i,t}$ attributable to its sector mean, $\mathrm{Var}_i(\bar{\tilde{z}}_{s(i),t})/\mathrm{Var}_i(\tilde{z}_{i,t})$, with the interquartile range across months. The "return comovement" columns report R^2 from univariate spanning regressions of the unadjusted HML return on the sector-wide and sector-adjusted HML returns, and the implied sector-wide share $R^2_{SW}/(R^2_{SW}+R^2_{SA})$. Sample: March 1987 to June 2024, 448 months.

An Exact Decomposition. The tercile portfolios of the three constructions hold different commodities, so the unadjusted return is not a combination of its two components, and the univariate R^2 values need not sum to the bivariate R^2 of the two. We therefore also form characteristic-managed portfolios with linear weights,

$$w_{i,t} = \frac{\widetilde{z}_{i,t}}{c_t} = \frac{\widetilde{S}_{i,t} + \widetilde{v}_{i,t}}{c_t}, \qquad c_t = \tfrac{1}{2}\sum_i |\widetilde{z}_{i,t}|, \tag{6}$$

so that the unadjusted portfolio is one dollar long and one dollar short, and the two components share its scaling. The unadjusted return is then the sum of the component returns, $r^U_{t+1} = r^{SW}_{t+1} + r^{SA}_{t+1}$, and its variance splits into $\theta^{SW} = [\mathrm{Var}(r^{SW}) + \mathrm{Cov}(r^{SW}, r^{SA})]/\mathrm{Var}(r^U)$ and $\theta^{SA} = 1 - \theta^{SW}$. These linear-weight portfolios serve the variance decomposition and its checks below, and the split of the momentum premium in Section 5.4; the tercile portfolios remain the test assets for the pricing tests of Section 3.3 and Section 4.

Panel B of Table 2 reports the decomposition. The sector-wide component accounts for a median 61% of the variance of the unadjusted return, with a larger share than the sector-adjusted component for 39 of the 46 characteristics. Under linear weights, the median univariate R^2 of the unadjusted return is 0.63 on the sector-wide and 0.43 on the sector-adjusted component, against 0.44 and 0.29 under the tercile construction, so the two

constructions order the components in the same way. Panel C shows that the linear-weight premia follow the same pattern as the tercile ones: the sector-adjusted mean is significant at the 5% level for 15 of the 46 characteristics and the sector-wide mean for 4, although the median sector-wide mean is the larger of the two. Section IA.C.11 of the Internet Appendix reports the decomposition by characteristic.

The asymmetry in Table 2 could arise from splitting any signal into group means and deviations from them. Over 1,000 random partitions of the 46 commodities into five groups of the sizes of the actual sectors, the median sector-wide variance share under linear weights is 0.11, the same as the between-group share of the variance of the ranked signals, against 0.61 and 0.32 under the economic sectors (Section IA.D.5 of the Internet Appendix). Counts over characteristics also overstate the number of independent observations, because many of the 46 characteristics are close variants of one another; grouped at an absolute return correlation of 0.5, they form 20 clusters (Section IA.D.6 of the Internet Appendix). On one representative per cluster, the sector-wide component has the larger tercile R^2 for 14 of the 20 and the larger linear-weight variance share for 13, so the direction of the asymmetry is unchanged, although with 20 observations statistical significance diminishes.

3.3 Pricing Content of the Two Components

The R^2 comparison does not settle which component prices the unadjusted return, since a portfolio can comove strongly with a component and still carry a significant intercept against it. Panel B of Figure 1 reports the absolute t-statistics of the intercepts of the univariate spanning regressions on each component and of the bivariate regression on both (Figure IA.8 of the Internet Appendix shows the intercepts). In the univariate regressions, the median absolute alpha is 1.93% annualized on the sector-wide component, with 11 of 46 characteristics significant at the 5% level, and 3.33% on the sector-adjusted component, with 10 significant, so neither component alone spans the unadjusted returns, and the sector-adjusted component leaves the larger alpha despite its smaller R^2. In the bivariate regressions, which include

both components, the median absolute alpha falls to 1.61%, with only 4 of 46 characteristics retaining an alpha significant at the 5% level (9 at the 10% level), and a median R^2 of 0.72.

We also run the spanning test in the reverse direction, regressing each sector-wide and sector-adjusted HML return on the unadjusted HML return of the same characteristic. This allows us to test whether the two components expand the investment opportunity set of the unadjusted portfolios. Table IA.3 in the Internet Appendix reports the results. The unadjusted portfolio largely spans its own components: median absolute alphas are 1.7% annualized for both, no sector-wide and 4 of 46 sector-adjusted intercepts are significant at the 5% level, and the joint Gibbons et al. (1989) test rejects for 4 of 46 characteristics. Combining the two components in a tangency portfolio raises the absolute Sharpe ratio compared to the unadjusted portfolio for 28 characteristics, from a median of 0.23 to 0.29.

4 Asset Pricing

We next examine whether the sector-wide and sector-adjusted channels carry distinct *priced* risk by estimating a linear stochastic discount factor from the cross-section of test asset returns.[7] In Section 4.1, we first extract latent factors from each set of test assets and compare their pricing performance within each construction and on a common cross-section. We then ask whether either channel earns a premium beyond standard commodity factor models in Section 4.2.

4.1 Priced Risk in Commodity Returns

Commodity futures markets are at best partially integrated with broader capital markets (Bessembinder, 1992; Acharya et al., 2013), so the investors who price commodity risk need not be those who price equities, and may differ across commodities and over time. A factor

[7] One could instead treat the decomposition as an input to commodity-level forecasting, for example, by selecting predictors through regularization or estimating time-varying loadings in rolling windows. Such methods optimize forecast accuracy, which is a distinct objective from compensation for risk, so we estimate the stochastic discount factor directly.

model imported from the equity literature would then fix the pricing kernel by assumption. We therefore estimate it with latent factor methods.

Latent factor methods also fit the sector structure of the data. Commodity futures curves share mostly sector-specific common factors (Karstanje et al., 2017), so characteristics built from futures prices carry sector structure, and the factors may differ across the three sets of test assets; latent factor methods estimate them separately for each. The observable commodity factors, namely momentum (Miffre and Rallis, 2007), basis (Erb and Harvey, 2006), and hedging pressure (Basu and Miffre, 2013), are also correlated with one another, so the premium of one cannot be attributed to it alone. The SDF we estimate is specific to commodity futures: it is extracted from the commodity cross-section and prices that cross-section, and we make no claim that it prices other asset classes such as equities.

We employ the three-pass procedure of Nucera et al. (2024), integrating the risk-premium principal component analysis (RP-PCA) of Lettau and Pelger (2020a,b) with Giglio and Xiu (2021). RP-PCA extracts factors that explain both the time-series variation and the cross-section of average returns, which matters when the test assets differ in their exposure to sector and commodity-specific risk.

Let X denote the $T \times N$ matrix of test portfolio returns. We assume K latent factors $F = [F_1, \ldots, F_K]$ capture the systematic component:

$$X = F\psi^\top + \varepsilon \tag{7}$$

where ψ is the $N \times K$ loading matrix and ε captures idiosyncratic risk.

Standard PCA extracts factors from the sample covariance matrix $\Sigma_{PCA} = \frac{1}{T}X^\top X - \overline{X}^\top \overline{X}$, maximizing explained time-series variation but ignoring return means, a limitation for tradable portfolios whose time-series averages proxy for risk premia (Cochrane, 2005).

Lettau and Pelger (2020a,b) address this by extracting factors from:

$$\Sigma_{RP} = \frac{1}{T} X^\top X + \omega \overline{X}^\top \overline{X} \tag{8}$$

where $\omega > 0$ overweights mean returns. In Lettau and Pelger (2020b), the resulting SDF prices the test assets with smaller errors and a higher maximal Sharpe ratio than the SDF based on standard PCA. The penalty $\omega = -1$ recovers standard PCA; $\omega = 0$ applies PCA to the second-moment matrix.

Asset Pricing Performance. Table 4 evaluates the SDF spanned by the first K factors, $K = 1, \ldots, 8$, at $\omega = -1$ and $\omega = 30$, with four diagnostics. Average idiosyncratic variance, $\overline{\sigma}^2_\epsilon = \sum_t \widehat{\mathrm{Var}}_n(\epsilon_{nt}) / \sum_t \widehat{\mathrm{Var}}_n(X_{nt})$, is the dispersion across test assets of the residuals of the first-pass regression $X_{nt} = \alpha_n + \widehat{F}_t \psi_n^\top + \epsilon_{nt}$ relative to that of returns. The root-mean-square pricing error is $\overline{RMS}_\alpha = \sqrt{\widehat{\alpha}^\top \widehat{\alpha}/N}$, annualized. The cross-sectional R^2 is from the regression of average returns $\overline{X}_n$ on the estimated loadings with an intercept. The Sharpe ratio is that of the tangency portfolio $\widehat{F}\widehat{b}_{MV}^\top$ with $\widehat{b}_{MV} = \widehat{\Sigma}_F^{-1} \widehat{\mu}_F$, annualized; it measures how close the estimated SDF comes to the efficient frontier (Hansen and Jagannathan, 1991), not the performance of a strategy net of trading costs.

In Table 4, explanatory power rises with each additional factor in all constructions, and the metrics stabilize beyond six factors, consistent with several risks carrying modest premia (Sakkas and Tessaromatis, 2020). With seven factors, cross-sectional R^2 values under RP-PCA are 99.45%, 98.48%, and 98.90% for the unadjusted, sector-wide, and sector-adjusted portfolios, each priced on its own test asset set, compared to 58.30%, 53.77%, and 30.17% under standard PCA.

Section IA.C.4 of the Internet Appendix shows that under standard PCA, only the first factor has a mean significantly different from zero at the 5% level in each construction. At $\omega = 30$, five of the seven factors of the baseline specification have significant means for

		Unadjusted				Sector-wide				Sector-adjusted			
RP weight	SDF	$\overline{\sigma}^2_\epsilon$	$\overline{RMS}_\alpha$	$R^2(\%)$	SR	$\overline{\sigma}^2_\epsilon$	$\overline{RMS}_\alpha$	$R^2(\%)$	SR	$\overline{\sigma}^2_\epsilon$	$\overline{RMS}_\alpha$	$R^2(\%)$	SR
$\omega = -1$	$\varphi(F_1)$	92.23	1.98	27.69	0.49	96.67	1.95	24.86	0.49	97.23	1.55	3.07	0.49
	$\varphi(F_{1-2})$	78.78	1.81	48.02	0.55	85.44	1.79	43.89	0.53	88.95	1.43	16.14	0.57
	$\varphi(F_{1-3})$	68.72	1.66	50.12	0.61	77.80	1.75	44.79	0.54	81.40	1.41	17.32	0.58
	$\varphi(F_{1-4})$	62.94	1.54	56.66	0.68	71.93	1.70	48.47	0.56	75.66	1.40	19.35	0.59
	$\varphi(F_{1-5})$	58.74	1.53	56.69	0.68	67.81	1.70	48.68	0.56	71.74	1.38	24.47	0.61
	$\varphi(F_{1-6})$	55.09	1.53	56.83	0.68	64.35	1.70	51.24	0.56	68.18	1.37	24.47	0.61
	$\varphi(F_{1-7})$	52.04	1.50	58.30	0.71	61.23	1.70	53.77	0.56	64.79	1.30	30.17	0.69
	$\varphi(F_{1-8})$	49.19	1.49	58.73	0.72	58.32	1.62	54.26	0.61	61.78	1.28	32.68	0.72
$\omega = 30$	$\varphi(F_1)$	92.14	1.97	83.01	0.51	96.56	1.94	83.41	0.52	97.25	1.55	79.99	0.50
	$\varphi(F_{1-2})$	79.40	1.75	89.91	0.73	85.56	1.78	83.88	0.58	89.59	1.47	93.77	0.90
	$\varphi(F_{1-3})$	69.11	1.67	95.23	0.80	77.94	1.74	87.27	0.62	81.99	1.46	94.89	0.90
	$\varphi(F_{1-4})$	63.36	1.57	98.09	1.01	72.10	1.70	89.12	0.67	76.66	1.44	95.64	1.08
	$\varphi(F_{1-5})$	59.65	1.56	98.43	1.19	68.57	1.65	95.76	0.88	72.89	1.14	98.59	1.42
	$\varphi(F_{1-6})$	56.54	1.23	99.38	1.49	65.42	1.31	98.27	1.09	69.14	1.11	98.90	1.45
	$\varphi(F_{1-7})$	52.97	1.20	99.45	1.50	61.96	1.31	98.48	1.09	65.58	1.11	98.90	1.45
	$\varphi(F_{1-8})$	50.15	1.11	99.57	1.56	58.85	1.31	98.71	1.09	62.58	1.05	99.06	1.50

Table 4: **Asset pricing performance.** Diagnostics for the SDF spanned by the first k RP-PCA factors, $\varphi(F_{1-k})$, $k = 1, \ldots, 8$, at penalty weights $\omega = -1$ (standard PCA) and $\omega = 30$: average idiosyncratic variance $\overline{\sigma}^2_\epsilon$ (percent), root-mean-square pricing error $\overline{RMS}_\alpha$ (annualized, percent), cross-sectional R^2 (percent), and the annualized Sharpe ratio of the tangency portfolio, defined in Section 4.1. Test assets are the 138 tercile portfolios formed from the 46 main-sample characteristics, built separately under each construction and priced by factors extracted from the same construction. Sample: 448 months, March 1987 to June 2024.

the unadjusted test assets, against four for the sector-adjusted and three for the sector-wide test assets. The fourth to sixth factors of the unadjusted test assets have normalized eigenvalues between 0.08 and 0.14, against 6.65 for the first, yet significant means and Sharpe ratios between 0.62 and 0.89 (see Table IA.5), consistent with risks that are priced without accounting for much of the comovement of returns (Lettau and Pelger, 2020b). RP-PCA therefore identifies weak but priced factors that standard PCA misses.

With seven factors, the maximal Sharpe ratio is 1.50 for the unadjusted, 1.45 for the sector-adjusted, and 1.09 for the sector-wide test assets. Unadjusted test assets combine sector-wide and sector-adjusted return content, which are nearly uncorrelated at the portfolio level (see Section 3), so a factor structure extracted from them spans both channels, whereas each set of decomposed test assets captures only one. The shortfall is nonetheless concentrated in the sector-wide test assets, and the sector-adjusted test assets attain close to the unadjusted Sharpe ratio.

	Unadjusted factors		Sector-wide factors		Sector-adjusted factors	
	$\overline{RMS}_\alpha$	R^2 (%)	$\overline{RMS}_\alpha$	R^2 (%)	$\overline{RMS}_\alpha$	R^2 (%)
$\varphi(F_{1-1})$	1.97	83.01	1.98	42.02	1.99	56.01
$\varphi(F_{1-2})$	1.75	89.91	1.77	52.34	1.66	56.47
$\varphi(F_{1-3})$	1.67	95.23	1.65	52.35	1.66	58.17
$\varphi(F_{1-4})$	1.57	98.09	1.53	58.63	1.59	62.27
$\varphi(F_{1-5})$	1.56	98.43	1.59	58.66	1.64	68.13
$\varphi(F_{1-6})$	1.23	99.38	1.61	59.74	1.69	69.41
$\varphi(F_{1-7})$	1.20	99.45	1.61	59.81	1.68	74.59
$\varphi(F_{1-8})$	1.11	99.57	1.60	60.49	1.61	74.59

Table 5: **Pricing performance on a common test asset space.** The table reports SDF pricing diagnostics for three sets of latent factors applied to the same cross-section of 138 unadjusted tercile portfolios. The first column block reproduces the pricing performance of latent factors extracted from unadjusted portfolios (Table 4, Unadjusted block). The second and third columns report the corresponding pricing performance when the latent factors are extracted from sector-wide and sector-adjusted portfolios, respectively. Factors are obtained by RP-PCA with penalty weight $\omega = 30$ and number of factors $k = 1, \ldots, 8$. For each cell, the table reports the root-mean-square pricing error $\overline{RMS}_\alpha$ (annualized in percent) from the first-pass time-series regressions and the cross-sectional R^2 from the second-pass regression $\overline{X}_n = \widehat{\psi}_n \gamma^\top + \alpha_n$. Sample: March 1987 to June 2024, 448 months.

Pricing on a Common Test Asset Space. Table 4 lets both the factors and the test assets vary across constructions, so cross-column comparisons mix differences in the extracted factors with differences in the cross-section being priced. Table 5 holds the test assets fixed at the 138 unadjusted tercile portfolios and prices them with factors extracted from each decomposed channel.

Table 5 reports the results for $\omega = 30$. With seven factors, sector-adjusted factors achieve a cross-sectional R^2 of 74.59% on the unadjusted test assets, compared with 59.81% for sector-wide factors, while pricing errors are comparable and slightly larger for the sector-adjusted factors ($\overline{RMS}_\alpha$ of 1.68 against 1.61). The ranking does not depend on the number of closely related characteristics. Section IA.D.6 of the Internet Appendix shows that when the test assets and factors are rebuilt from one representative of each of 20 clusters of characteristics, sector-adjusted factors price the unadjusted portfolios better in 98% of 2,000 random choices of representatives.

Both sets of decomposed factors fall short of the native unadjusted factors, which achieve

$R^2 = 99.45\%$ with $\overline{RMS}_\alpha = 1.20$ at $K = 7$. Part of the gap is mechanical, since factors extracted from the unadjusted test assets are optimized for that cross-section. Yet sector-adjusted factors recover roughly three-quarters of the native R^2, and sector-wide factors roughly three-fifths. Beyond $K = 3$, the R^2 gap between the two decomposed channels widens with K, from 3.6 percentage points at $K = 4$ to 14.8 at $K = 7$, and is 14.1 at $K = 8$. Factors extracted from the sector-adjusted portfolios therefore price the unadjusted cross-section better than factors extracted from the sector-wide portfolios, an asymmetry that the spanning regressions of Section 3, in which neither component alone spans the unadjusted returns, do not show.

Latent Factors and the Observable Portfolios. Section IA.C.6 of the Internet Appendix relates the latent factors to the observable characteristic-managed portfolios. Market risk loads on the first factor in all three constructions and momentum on the second, most strongly for unadjusted portfolios, while the higher-order factors load on different combinations of characteristics once sector content is removed. The observable portfolios do not fully span the latent factors, since their total R^2 on the factors remains well below one in all three constructions. The gap could reflect priced exposure that the factors miss or variation in the observable portfolios unrelated to the SDF, and the weak-factor test of Giglio and Xiu (2021) distinguishes the two. With one factor, the null that a characteristic is weakly exposed to the factors is not rejected at the 5% level for 19, 25, and 31 of the 46 characteristics in the unadjusted, sector-wide, and sector-adjusted constructions, but once $K \geq 2$ it is not rejected for at most 11, consistent with the second explanation (see Section IA.C.7 of the Internet Appendix).

4.2 Observable Commodity Factor Models

We consider two benchmark specifications with observable factors. The 4-factor specification consists of HML portfolios formed on momentum, basis, and basis-momentum, the charac-

teristics behind the factors of Szymanowska et al. (2014), Bakshi et al. (2019), and Boons and Porras Prado (2019), together with a commodity market factor calculated as the equal-weighted average excess return of all commodities with a return in the month. The 6-factor specification adds HML portfolios on value and hedging pressure, matching the five predictors of the panel regressions summarized in Section 2.3. All benchmark HML factors are constructed from the unadjusted characteristics. Section IA.D.7 of the Internet Appendix reports the results for the 3- and 5-factor benchmarks without the market factor.

For each of the 46 characteristics except those whose HML portfolios form the benchmark (momentum, basis, and basis-momentum in the 4-factor specification, and also value and hedging pressure in the 6-factor specification), we regress its HML return under each construction on the benchmark factors,

$$r^D_{c,t} = \alpha^D_c + \boldsymbol{\beta}^{D\prime}_c \boldsymbol{f}_t + \varepsilon^D_{c,t}, \qquad D \in \{\text{unadj.}, \text{SW}, \text{SA}\}, \tag{9}$$

where $r^D_{c,t}$ is the HML return of characteristic c under construction D and $\boldsymbol{f}_t$ is the vector of benchmark factors, which are the same for the three constructions. This leaves 43 test portfolios for the 4-factor and 41 for the 6-factor specification, and the GRS test is applied to the alphas of each construction jointly. The GRS statistic uses the maximum-likelihood residual covariance matrix and its finite-sample $F(N, T-N-K)$ distribution.

Table 6 reports the results. The unadjusted HML portfolios reject both benchmarks at the 5% level under the GRS test (GRS p-values of 0.011 and 0.049), with median R^2 values of 27.2% and 29.2%. This is consistent with the broader commodity asset pricing literature: a small set of standard factors does not span the full cross-section of characteristic-based premia (Sakkas and Tessaromatis, 2020). The sector-wide HML portfolios reject neither benchmark (GRS p-values of 0.092 and 0.292), whereas the sector-adjusted HML portfolios reject both at the 1% level (GRS p-values of 0.000 and 0.002), despite exhibiting the lowest comovement with the benchmark factors (median R^2 of 7.6% and 10.0%, against 18.4% and

	4-factor benchmark			6-factor benchmark		
	Unadj.	Sector-wide	Sector-adj.	Unadj.	Sector-wide	Sector-adj.
Median $\lvert\alpha\rvert$ (%)	1.96	2.04	1.78	1.53	1.92	2.01
Mean $\lvert\alpha\rvert$ (%)	2.48	2.54	2.10	2.05	2.21	2.07
# sign at 5%	6	5	8	4	2	8
Median R^2 (%)	27.2	18.4	7.6	29.2	20.6	10.0
GRS statistic	1.60	1.32	2.03	1.42	1.12	1.86
GRS p-value	0.011	0.092	0.000	0.049	0.292	0.002
GRS, SW + SA	1.54 ($p = 0.004$)			1.40 ($p = 0.020$)		

Table 6: **Alphas of HML portfolios against benchmark commodity factor models.** Time-series alphas of the HML portfolios of the characteristics not used in the benchmark (43 for the 4-factor and 41 for the 6-factor benchmark of Section 4.2), with the unadjusted, sector-wide, and sector-adjusted HML portfolios as test assets in turn. Median $\lvert\alpha\rvert$ and mean $\lvert\alpha\rvert$ are cross-characteristic statistics of the absolute intercepts, annualized in percent; median R^2 is the cross-characteristic median of the time-series R^2. GRS is the Gibbons et al. (1989) joint test of zero alphas; the final row reports it for the pooled panel of sector-wide and sector-adjusted portfolios. Sample: March 1987 to June 2024, $T = 448$ months.

20.6% for the sector-wide portfolios).

The higher comovement of the sector-wide portfolios with the benchmark factors is consistent with factors built from unadjusted characteristics reflecting mainly the sector-wide channel. The rejection by the sector-adjusted portfolios does not, however, depend on how the benchmark factors are constructed: benchmarks built from the sector-adjusted characteristics themselves are rejected with GRS p-values of at most 0.001 (Section IA.D.8 of the Internet Appendix). The sector-adjusted premia are therefore not spanned by a small number of characteristic-sorted factors of any construction.

The final row of Table 6 considers the two channels together: the pooled panel of sector-wide and sector-adjusted portfolios rejects spanning by both benchmarks (GRS p-values of 0.004 and 0.020). The joint rejection comes from the sector-adjusted panel, which rejects on its own with lower p-values, and adding the sector-wide portfolios weakens the evidence against the benchmarks.

The same pattern holds when considering one representative of each of the 20 clusters

of closely related characteristics, where the sector-adjusted portfolios reject the 4-factor benchmark at the 5% level ($p = 0.034$), the 6-factor benchmark at the 10% level ($p = 0.086$), and the sector-wide portfolios reject neither (p-values of 0.775 and 0.945; Section IA.D.6 of the Internet Appendix).

Individual Commodity Returns. As a complementary test, we ask whether the decomposition between sector-wide and sector-adjusted components changes the exposures of individual commodity futures to characteristic-sorted factors. To this end, we regress excess returns for each commodity on the five HML portfolios of the panel regressions of Section 2.3, constructed in turn from each of the three versions of the characteristics. Section IA.C.12 of the Internet Appendix reports the results.[8]

How many commodities load on the factors changes little across constructions: at least one loading is significant for 82%, 76%, and 73% of commodities under the unadjusted, sector-wide, and sector-adjusted constructions, with median R^2 values of 4.9%, 5.9%, and 4.5%. Which factors they load on changes more: hedging pressure is significant for 53% of commodities under the sector-wide construction, and for 11% under the sector-adjusted construction, and basis moves the other way, from 27% to 64%. A Benjamini and Hochberg (1995) correction across all commodity–factor pairs of a construction lowers the shares (76%, 62%, and 58% with at least one significant loading; hedging pressure 47% against 2%, basis 18% against 56%) but leaves the pattern unchanged. The exposure of individual commodities to hedging pressure therefore runs mainly through their sector-level component. Section 5 examines the extent to which position data confirm this result.

5 Economic Mechanisms

To examine why the two channels are priced differently, we test the implications of the equilibrium model of Appendix A, in which commercial hedgers with net physical exposure

[8] Commodity futures with fewer than 60 monthly observations are excluded.

of either sign trade with speculators. We then examine momentum, for which we test three explanations from the literature and one from the model.

The model has two kinds of implications. The first concerns the pattern of risk transfer: speculators hold the opposite side of hedgers' positions at both levels, the two premia take the signs and ordering of hedgers' exposure, and the two premia diverge over the cycle only if the within-sector dispersion of hedging exposure changes. The data are consistent with all three, although the slopes on positions that give the signs and ordering are not significant. The second concerns the source of the premia, which in the model is the hedgers' net exposure. With exposure measured from positions, the data are not consistent with it: the portfolio built from hedgers' sector-adjusted positions does not price the sector-adjusted portfolios. The source of the sector-adjusted premium therefore remains open, and none of the four explanations tested in Section 5.4 accounts for the sector-adjusted component of momentum, the one characteristic whose premium divides evenly between the two channels.

5.1 Hedging Demand and Trader Positions

We measure positions with the CFTC Disaggregated Commitments of Traders (DCOT) reports. *Producers and merchants*, the Producer/Merchant/Processor/User category, are commercial traders hedging physical exposure; the category nets long hedging by processors and users against short hedging by producers, so its net position corresponds to the hedgers' net futures position of the model. In the CFTC classification, *swap dealers* intermediate OTC derivatives with commercial clients and hedge the resulting exposure in futures, and *managed money* comprises commodity trading advisors and hedge funds.

The DCOT classifications begin in June 2006 and cover 32 of the 46 commodities, so the positioning evidence spans roughly half of the return sample, a period in which the financialization of commodity markets changed the composition of financial participants (Cheng and Xiong, 2014; Tang and Xiong, 2012).[9] The evidence is descriptive and does not

[9]The DCOT data are available from June 2006 to December 2024. The remaining commodities are not

	Swap Dealers	Managed Money	Producers
Panel A: Variance decomposition			
Sector-wide share (%)	20.9 (22.4)	24.7 (23.1)	25.9 (25.8)
Sector-adjusted share (%)	79.1 (77.6)	75.3 (76.9)	74.1 (74.2)
Sector-wide share, gross short (%)	–	–	12.7 (11.6)
Panel B: Correlation with net producer positions			
Sector-wide (ρ^{SW})	−0.074	−0.651	–
Sector-adjusted (ρ^{SA})	−0.351	−0.598	–
Panel C: Correlation with gross short producer positions			
Sector-wide (ρ^{SW})	−0.408	−0.407	–
Sector-adjusted (ρ^{SA})	−0.034	−0.358	–

Table 7: **Trader positions by component.** Net positions over open interest by CFTC trader category, decomposed as in Equation (10). Panel A reports the sector-wide share of the cross-sectional variance of positions, averaged over months, with the median month in parentheses; the third row measures the producer leg by gross short rather than net positions. Panels B and C report the correlation between a speculator category's component and the producers' component of the same type, pooled across commodity-months, with the producer leg measured by net positions (Panel B) and by gross short positions entered with a minus sign (Panel C), so that a negative correlation means the speculator holds the opposite side of producer hedging. A dash marks a cell to which the measure does not apply. Sample: June 2006 to December 2024, 32 commodities, 6,813 commodity-months.

establish the direction of causality between positions and premia.

For trader type k in month t, commodity i has net position normalized by open interest $NP_{i,t}^{(k)}$, which we decompose as the characteristics are decomposed in Equations (4)–(5):

$$NP_{i,t}^{(k)} = \underbrace{\frac{1}{n_{s(i)}} \sum_{j \in s(i)} NP_{j,t}^{(k)}}_{NP_{i,t}^{(k),SW}} + \underbrace{NP_{i,t}^{(k)} - \frac{1}{n_{s(i)}} \sum_{j \in s(i)} NP_{j,t}^{(k)}}_{NP_{i,t}^{(k),SA}}, \tag{10}$$

where $n_{s(i)}$ is the number of commodities of sector $s(i)$ reported in month t. Table 7 reports the sector-wide share of the cross-sectional variance of each trader type's positions (Panel A) and the pooled correlation of each speculator category's sector-wide and sector-adjusted components with the corresponding components of the producers' position (Panels B and C).

in our DCOT sample. Most of them trade on non-US exchanges, among them the LME-traded metals (lead, nickel, tin, zinc), iron ore, and rubber; for the others, which include propane, pork bellies, and cobalt, the DCOT reports cover the traded contract for most of the period.

In the model, speculators take the other side of hedgers' net positions at both the sector-wide and the sector-adjusted level (Proposition 3). Managed money does so: its positions offset producers' net positions at the sector-wide level ($\rho^{SW} = -0.651$) and at the sector-adjusted level ($\rho^{SA} = -0.598$) with similar magnitudes.

The model also allows for a second speculator clientele whose positions are neutral within each sector, so that it holds no sector-wide position and takes part in pricing only the sector-adjusted component. If such a clientele were present, the sector-wide share of position variance would be larger for the unrestricted speculators than for hedgers; with a single speculator type, the two shares are equal. In the data, the shares are 24.7% for managed money and 25.9% for producers, and their monthly difference averages -1.2 percentage points ($t = -0.72$; Table IA.23 in Section IA.D.9 of the Internet Appendix).

Swap dealers do not fit the sector-neutral description either: they offset producers' net positions mainly within sectors ($\rho^{SA} = -0.351$ against $\rho^{SW} = -0.074$), but the sector-wide component accounts for 20.9% of the variance of their positions (Panel A), and they offset the sector-wide component of producers' gross short positions ($\rho^{SW} = -0.408$ against $\rho^{SA} = -0.034$, Panel C). Their positions may also reflect the commodity composition of their OTC client books, which the positions data cannot separate from a choice of which risk to bear. We therefore take the single speculator type as the baseline of Appendix A.

5.2 Premia and Hedging Positions

In the model, each premium component is proportional to the matching component of hedgers' net physical exposure, and hedgers' net futures position is a negative multiple of that exposure (Proposition 1), so a commodity's premium should fall with hedgers' net long position at the sector-wide and at the sector-adjusted level, and by more per unit at the sector-wide level. We estimate

$$r_{i,t+1} = a + b_W \, NP_{i,t}^{(P),SW} + b_A \, NP_{i,t}^{(P),SA} + e_{i,t+1}, \tag{11}$$

Panel A: Returns on hedging positions			
	b_W	b_A	$b_W - b_A$
Coefficient	-1.394	-0.276	-1.118
	(-0.80)	(-0.60)	(-0.60)
Panel B: Within-sector dispersion in recessions			
	Median coefficient	Negative, significant	Positive, significant
Characteristics, dispersion	0.118	3 of 46	13 of 46
Characteristics, within-sector share	-0.099	16 of 46	2 of 46
Hedging positions, dispersion	-0.327 (-3.51)		

Table 8: **Tests of the model's pricing implication and dispersion assumption.** Panel A reports b_W and b_A from Equation (11), the regression of monthly commodity returns (percent) on the sector-wide and sector-adjusted components of net Producer/Merchant/Processor/User positions over open interest, with Driscoll–Kraay t-statistics (5 lags) in parentheses; 6,540 commodity-months, 32 commodities, June 2006 to May 2024. Panel B reports the recession coefficient from the regression of the log monthly cross-sectional variance of the sector-adjusted component on a recession indicator (FRED USRECD) and a linear trend: for the 46 characteristics, on their own scale, the median coefficient and the number of characteristics with a coefficient significantly negative or positive at the 5% level (Newey–West, 5 lags), for the dispersion and for the within-sector share of dispersion; for producers' net positions, the coefficient with its t-statistic in parentheses (223 months, 20 in recession).

where $NP^{(P)}$ is the producers' net position over open interest, with Driscoll–Kraay standard errors, which allow for correlation across commodities within a month. Panel A of Table 8 reports the estimates. Both coefficients are negative, $b_W = -1.394$ and $b_A = -0.276$ (returns in percent per month), and the sector-wide coefficient is the larger in magnitude, as the model predicts, but none of the three estimates, including the difference $b_W - b_A = -1.118$, is statistically significant (t-statistics of -0.80, -0.60 and -0.60).

Hedging pressure, hedgers' net short position over their gross positions, is available over 1987–2024 against 2006–2024 for the DCOT positions, and because it measures net short rather than net long positions, the predicted slopes are positive. Both slopes are positive, and the sector-wide slope is the larger (1.27 and 0.70 percent a month, t-statistics of 1.43 and 1.87), but neither is significant at the 5% level, and the 95% confidence interval for their difference, from -1.31 to 2.43, contains both zero and values more than four times the point estimate of 0.56 (see Table IA.24).

The model also implies which portfolio should price the sector-adjusted portfolios. Be-

cause sector-adjusted premia are proportional to hedgers' sector-adjusted exposure and commodity shocks have equal variance within a sector (Proposition 1 and assumption A1), the portfolio that holds each commodity in proportion to that exposure is mean-variance efficient within sectors, and the 46 sector-adjusted HML portfolios should have zero alphas against it. Hedgers' exposure is not observed, but their futures position is a negative multiple of it (Proposition 3), so we form the portfolio with weights equal to minus the sector-adjusted component of producers' net positions. It earns 1.55% a year ($t = 0.59$) and explains a median 1.6% of the variance of the sector-adjusted HML portfolios, and the GRS test rejects zero alphas ($p = 0.004$). The same portfolio formed from hedging pressure over 1987–2024 earns 3.90% a year ($t = 1.88$) and is also rejected ($p < 0.001$; Table IA.23). Hedging exposure, as measured in the positions data, therefore does not significantly account for the pricing content of the sector-adjusted portfolios.

5.3 Risk Premia over the Business Cycle

With a single speculator type, a fall in speculators' risk-bearing capacity raises both premia in the same proportion (Equation (A.6) in Appendix A), so the two components can diverge across the cycle only through hedging exposure. A decline in sector-adjusted premia in recessions requires the within-sector dispersion of hedging exposure to compress, which the model takes as an assumption.

Figure IA.16 in Section IA.D.3 of the Internet Appendix plots the distribution across characteristics of the change in annualized Sharpe ratio from expansions to recessions, with a month classified as a recession if the FRED indicator USRECD equals one on any day of that month. The sector-wide distribution has a positive median, $\Delta SR = +0.11$, and the sector-adjusted distribution lies to the left, with a median $\Delta SR = -0.31$ and 65% of characteristics declining, so the two channels move in opposite directions across the cycle. Unadjusted portfolios also improve in recessions (median $\Delta SR = +0.19$).

Table IA.15 in Section IA.D.3 of the Internet Appendix reports the SDF diagnostics of Table 4 separately in expansions and recessions, with the factors and the tangency weights held at their full-sample estimates (Kelly et al., 2019), which avoids the mechanical inflation of in-sample Sharpe ratios in a 36-month subsample. With seven factors, the cross-sectional R^2 is 97.24%, 91.84%, and 95.50% for unadjusted, sector-wide, and sector-adjusted portfolios in expansions and 52.95%, 44.39%, and 24.81% in recessions, so the sector-adjusted portfolios lose most of their pricing fit in recessions, and pricing errors rise fourfold to fivefold in every construction. The tangency Sharpe ratio falls in recessions in all three constructions, from 1.53 to 1.29, from 1.26 to 0.24, and from 1.49 to 0.94.

Panel B of Table 8 tests the dispersion assumption by regressing the log of the monthly cross-sectional variance of the sector-adjusted component of producers' net positions, and of each characteristic, on a recession indicator and a linear trend. The within-sector dispersion of producers' net positions falls by 0.327 in recessions relative to expansions ($t = -3.51$), although the positions data contain only 20 recession months. The within-sector dispersion of the characteristics rises in recessions (median coefficient 0.118, with significant increases for 13 of the 46 and decreases for 3), but as a share of total cross-sectional dispersion it falls (median -0.099, with significant decreases for 16 and increases for 2), because sector-wide dispersion rises more. Hedging positions, the model's own measure of exposure, therefore compresses as the assumption requires.

Figure IA.18 reports the average net position of each trader category in expansions and recessions. Producers increase their net short positions from -21.7% to -26.6% of open interest, swap dealers expand their net long positions from 10.8% to 15.4%, and managed money positions show no significant change (t-statistics of the expansion-minus-recession difference of 3.02, -11.72, and -0.49). With hedging exposure fixed, the model implies that hedgers reduce their positions when speculators' capacity falls, so the larger producer short positions in recessions are consistent with the model only if hedging exposure rises in recessions.

5.4 Momentum Across the Two Components

Under linear weights, the sector-wide and sector-adjusted components of twelve-month momentum earn 5.21% and 5.22% a year. Under the tercile construction, the sector-adjusted component earns 5.41% a year with an appraisal ratio of 0.42, against 0.31 for the sector-wide component, although the sector-wide component accounts for 68% of the variance of momentum returns (see Table 3). The momentum premium therefore divides almost evenly between the two channels, whereas a large share of equity momentum is accounted for by industry effects (Moskowitz and Grinblatt, 1999).

The even split of the premium is consistent with the small size of commodity sectors. Each sector contains between seven and twelve contracts whose returns are only moderately correlated (average within-sector correlations between 0.10 and 0.52), and the between-sector component accounts for a median of 32% of the cross-sectional variance of a characteristic, so a within-sector ranking retains most of the information in a signal.

Among the explanations of momentum in the literature, gradual diffusion of information across related commodities (Hong and Stein, 1999; Hong et al., 2007) implies that sector-adjusted momentum returns predict subsequent sector-level returns. Panel A of Table 9 tests this in two ways. At the portfolio level, we regress the sector-wide tercile HML return h months ahead on the current sector-adjusted and sector-wide returns,

$$r^{SW}_{t+h} = a_h + b_h\, r^{SA}_t + c_h\, r^{SW}_t + e_{t+h}. \tag{12}$$

At the sector level, we split the sector-adjusted linear-weight momentum return into the part earned on the commodities of each sector, $x_{s,t}$, and regress the equal-weighted return of sector s h months ahead on it,

$$R_{s,t+h} = a_h + b_h\, x_{s,t} + c_h\, R_{s,t} + \mu_s + e_{s,t+h}, \tag{13}$$

Panel A: Lead-lag, twelve-month momentum					
Horizon h (months)	1	2	3	6	12
Portfolio level, b_h	0.042	0.135	0.012	-0.196	-0.052
	(0.43)	(1.69)	(0.13)	(-1.75)	(-0.62)
Sector panel, b_h	0.003	-0.123	-0.077	0.021	0.084
	(0.04)	(-1.26)	(-0.72)	(0.23)	(0.61)
Panel B: Cost of arbitrage, twelve-month momentum					
Proxy	γ^{SA}	γ^{SW}	$\gamma^{SA} - \gamma^{SW}$	Observations	
Amihud illiquidity	0.215	0.045	0.168	12,488	
	(0.80)	(0.13)	(0.71)	1995–2024	
Zero trading days	0.444	0.142	0.310	13,031	
	(1.03)	(0.42)	(1.31)	1995–2024	
Inverse open interest	0.378	-0.085	0.465	13,544	
	(1.35)	(-0.22)	(1.15)	1987–2024	
Basis (storage cost)	0.054	-0.087	0.146	17,012	
	(0.20)	(-0.34)	(0.38)	1987–2024	

Table 9: **Momentum across the two channels: diffusion and the cost of arbitrage.** Panel A reports b_h from Equation (12) at the portfolio level, with Newey–West t-statistics ($\max(h, 5)$ lags), and from its sector-panel counterpart, with Driscoll–Kraay t-statistics ($\max(h, 5)$ lags). Panel B reports the interaction coefficient γ from Equation (14) for the sector-adjusted (γ^{SA}) and the sector-wide (γ^{SW}) momentum signal, and their difference from the regression with both signals, with higher values of the proxy meaning costlier to arbitrage. Coefficients in percent per month; t-statistics, clustered by commodity and by month, in parentheses. Sample: March 1987 to June 2024; illiquidity and zero trading days from the mid-1990s.

with sector fixed effects μ_s. Under gradual diffusion, $b_h > 0$ in both regressions. For twelve-month momentum, the sector-adjusted return does not predict the next month's sector-wide return ($b_1 = 0.042$, $t = 0.43$), and in the sector panel, a sector's contribution to the sector-adjusted return does not predict its return at any of the horizons reported, from one to twelve months.

Slow-moving capital (Duffie, 2010) implies that sector-adjusted momentum profits are larger among commodities that are costlier to arbitrage. Panel B tests this in the panel regression

$$r_{i,t+1} = \beta\, m_{i,t} + \gamma\, m_{i,t} x_{i,t} + \delta\, x_{i,t} + \mu_{s(i)} + \tau_t + e_{i,t+1}, \tag{14}$$

where $m_{i,t}$ is the sector-adjusted or the sector-wide twelve-month momentum signal, $x_{i,t}$

is one of four cost-of-arbitrage proxies standardized across commodities each month (Amihud illiquidity, the fraction of zero-trading days, inverse open interest, and the basis), and $\mu_{s(i)}$ and τ_t are sector and month fixed effects. The interaction of sector-adjusted momentum with each proxy is positive but not significant (t-statistics between 0.20 and 1.35), as is its difference from the interaction with sector-wide momentum. Section IA.D.4 of the Internet Appendix extends the lead-lag test to the linear-weight construction and to the eleven momentum and residual-momentum characteristics, with the same result, and applies a correction for testing across the four cost-of-arbitrage proxies.

Informational frictions imply that momentum profits are larger among commodities whose prices incorporate information slowly. Two proxies for slow processing, price delay (Hou and Moskowitz, 2005) and information continuity (Da et al., 2014), replace the cost-of-arbitrage proxy in Equation (14); neither interaction with sector-adjusted momentum is significant (t-statistics of 1.02 and -0.80). The model of Appendix A offers a fourth explanation: sector-adjusted premia are proportional to commodity-specific hedging exposure (Proposition 1), so if that exposure persists, sector-adjusted momentum may proxy for it. Adding the sector-adjusted component of hedging pressure to a panel regression of returns on sector-adjusted momentum lowers the momentum coefficient by 12%, from 0.575 to 0.508 percent a month, and the hedging-pressure component is not significant ($t = 1.14$; Section IA.D.10 of the Internet Appendix). None of these explanations accounts for the sector-adjusted momentum premium in our sample.

6 Extensions and Robustness

We examine the patterns of Sections 3 and 4 out of sample, over alternative sample periods, and with an alternative method of factor extraction, keeping the RP-PCA specification with $\omega = 30$ wherever RP-PCA is used, and we apply the decomposition to calendar-spread returns.

Out-of-Sample Performance. For each month t, we estimate RP-PCA on an expanding window with an initial length of 120 months and price the test assets at $t+1$ with the factors, loadings, and tangency weights estimated through t. Table IA.12 in Section IA.D.1 of the Internet Appendix reports the results. Unadjusted portfolios retain the most pricing power out of sample ($R^2 = 56.07\%$ with seven factors, still rising at eight), whereas both decomposed channels degrade more and to a similar degree ($R^2 = 37.46\%$ for sector-wide and 38.76% for sector-adjusted), and neither improves materially beyond seven factors. The first sector-wide factor prices a fifth of the cross-sectional dispersion in average returns on its own ($R^2 = 21.65\%$), whereas the first sector-adjusted factor prices little of it ($R^2 = 4.66\%$) and the sector-adjusted channel reaches its final level only with six factors. The sector-adjusted SDF nonetheless has the highest out-of-sample Sharpe ratio (0.61 at $K = 7$, against 0.52 for unadjusted and 0.09 for sector-wide) and the lowest pricing errors, so the pricing information of the sector-adjusted component is more stable out of sample than that of the sector-wide component by these two measures, although the two reach similar cross-sectional R^2.

Alternative Sample Periods. We repeat the analysis on an extended sample (August 1969 to June 2024, 40 characteristics) and a recent sample (January 1996 to June 2024, 52 characteristics), reported in Tables IA.14 and IA.13, respectively, in Section IA.D.2 of the Internet Appendix. The extended sample achieves an R^2 of 99.20% with six factors for unadjusted portfolios; the recent sample achieves 99.54%. In both samples, as in the main sample, sector-adjusted portfolios have the lowest pricing errors for every K, and the unadjusted and sector-adjusted SDFs attain similar maximal Sharpe ratios, both above that of the sector-wide SDF (1.20 and 1.23 against 0.78 in the extended sample, and 1.61 and 1.65 against 1.21 in the recent sample, at $K = 7$).

Alternative Factor Extraction. PCA-based estimators, RP-PCA included, can be inconsistent when only a small fraction of the test assets is exposed to a factor. Supervised PCA (SPCA; Giglio et al., 2025) selects, for each candidate factor, the test assets most correlated

with it before extracting principal components (Section IA.C.8 of the Internet Appendix). The first two SPCA factors dominate in every construction, as under RP-PCA, with the first factor explaining the least for the sector-wide portfolios (Figure IA.12). Because the SPCA factors are portfolios, they attain the maximal Sharpe ratio to the extent that they capture the SDF (Giglio et al., 2025, Section I.C.3), so the gap between their out-of-sample Sharpe ratio and that of the observable characteristic-managed portfolios measures how far the latter fall short of spanning it.

Figure IA.15 in Section IA.C.8 reports this comparison. At $K = 7$, the median Sharpe ratio spread is 0.25 for unadjusted and sector-wide portfolios and 0.34 for sector-adjusted portfolios, and SPCA delivers the higher Sharpe ratio for 38, 42, and 43 of the 46 characteristics; the exceptions are characteristics whose observable factors already earn high out-of-sample Sharpe ratios, such as twelve-month momentum. For the typical characteristic the observable factor earns little out of sample, with median Sharpe ratios of 0.17, 0.01, and 0.01, against 0.40, 0.28, and 0.42 for the SPCA-implied SDF portfolio. The spread is largest for the sector-adjusted portfolios, consistent with the rejection of standard commodity benchmarks being concentrated in the sector-adjusted channel (Table 6).

Calendar-Spread Returns. Applied to calendar-spread returns, the return on a long second-nearby position financed by a short first-nearby position, the linear-weight decomposition of Section 3.2 assigns a median 39% of the variance to the sector-wide component, against 59% for nearby returns on the same commodity-months, so calendar-spread returns comove mainly within sectors. The spread premia earned by sorting on the basis and on the relative basis are sector-adjusted, although across all characteristics the sector-wide spread mean is significant more often than the sector-adjusted one (18 against 5 of 46; Section IA.D.11 of the Internet Appendix).

7 Conclusion

We decompose commodity characteristics into orthogonal sector-wide and sector-adjusted components and ask how far characteristic-based risk premia reflect sector exposures and how far commodity-specific information. Under linear weights, the sector-wide component accounts for a median 61% of the variance of the unadjusted return, although it carries less than a third of the signal variance, and random groupings of commodities do not produce this asymmetry. Neither component alone prices the unadjusted returns. Latent factors extracted from the sector-adjusted portfolios price the unadjusted cross-section better than those extracted from the sector-wide portfolios (cross-sectional R^2 of 74.6% against 59.8%), and sector-adjusted HML portfolios reject the standard commodity factor benchmarks, with and without the commodity market factor, whereas sector-wide portfolios reject neither.

The median Sharpe ratio of sector-wide HML portfolios rises in NBER-dated recessions, and that of sector-adjusted portfolios falls. Managed money offsets producer hedging demand at both the sector and the commodity level. We interpret this evidence through an equilibrium model in which hedgers trade with speculators and the sector structure of payoff risk separates the two premia. The data are consistent with the model's pattern of risk transfer in positions, in the signs and ordering of the premia, although the slopes are not significant, and in the compression of within-sector hedging dispersion in recessions, but not with hedgers' net exposure as the priced risk: measured from positions, it does not price the sector-adjusted portfolios. The source of the sector-adjusted premium remains open.

The commodity momentum premium divides almost evenly between the two channels, whereas a large share of equity momentum is industry momentum, and none of the four explanations we test, gradual diffusion of information, slow-moving capital, informational frictions, and persistent hedging imbalances, accounts for its sector-adjusted component.

The pricing content of commodity characteristics is therefore more local than the sec-

tor structure of their returns would indicate. Factor models that include the sector-adjusted component may describe commodity risk premia more completely, and future work that identifies what drives the sector-adjusted premium will be addressing the central open question this paper leaves.

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Appendix for:

The Sector Anatomy of Commodity Risk Premia

A A Model of Risk Transfer Within and Across Sectors

This appendix derives the sector decomposition of risk premia in an equilibrium model in which premia compensate speculators for absorbing the net hedging demand of commercial traders (Keynes, 1930; Hirshleifer, 1988; De Roon et al., 2000). Premia separate into a sector-wide component that depends only on sector-wide hedging exposure and a sector-adjusted component that depends only on sector-adjusted exposure. Sector-wide exposure is priced more per unit, because sector risk is common to the commodities of a sector.

The separation follows from the factor structure of payoffs, and it holds whether or not some speculators can hold only positions that are neutral within each sector. We allow for such a clientele, which would contribute to pricing the sector-adjusted component. The positions data do not support it (Section 5.1), so our baseline has a single speculator type. Section A.6 separates what is assumed from what is derived, and the proofs are in Section A.7.

A.1 Setup

There are N commodities indexed by i, partitioned into S sectors; $s(i)$ is the sector of commodity i and n_s the number of commodities in sector s. Trading takes place at $t = 0$ and payoffs are realized at $t = 1$. The spot price of commodity i at $t = 1$ is

$$P_i = \bar{P}_i + \eta_{s(i)} + \epsilon_i, \tag{A.1}$$

with sector shocks $\eta_s \sim \mathcal{N}(0, \sigma_\eta^2)$ and commodity shocks $\epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2)$, all mutually independent. Let $\mathbf{B}$ be the $N \times S$ matrix with $B_{is} = 1$ if $s(i) = s$ and zero otherwise. The

covariance matrix of payoffs is

$$\boldsymbol{\Sigma} = \sigma_\eta^2 \, \mathbf{B}\mathbf{B}^\top + \sigma_\epsilon^2 \, \mathbf{I}_N. \tag{A.2}$$

A futures contract on commodity i trades at price F_i and pays $P_i - F_i$ per long contract. The risk premium is $\pi_i = \mathbb{E}_0[P_i] - F_i$, and $\boldsymbol{\pi}$ collects them.

The model uses two subspaces of $\mathbb{R}^N$. The sector-wide subspace $\mathcal{W}$ is spanned by the columns of $\mathbf{B}$ and contains the vectors that are constant within each sector. The sector-adjusted subspace $\mathcal{V} = \{\mathbf{v} : \mathbf{B}^\top \mathbf{v} = \mathbf{0}\}$ is its orthogonal complement and contains the vectors that sum to zero within each sector. Let $\mathbf{M}_W = \mathbf{B}(\mathbf{B}^\top \mathbf{B})^{-1}\mathbf{B}^\top$ be the projection on $\mathcal{W}$, which replaces each element of a vector by its sector mean, and $\mathbf{M}_V = \mathbf{I}_N - \mathbf{M}_W$ the projection on $\mathcal{V}$. For any vector $\mathbf{a} \in \mathbb{R}^N$ we write $\mathbf{a}^W = \mathbf{M}_W \mathbf{a}$ and $\mathbf{a}^A = \mathbf{M}_V \mathbf{a}$; element i of $\mathbf{a}^W$ is the mean of $\mathbf{a}$ over sector $s(i)$, denoted $a^W_{s(i)}$, and $a_i = a^W_{s(i)} + a^A_i$. The decomposition is that of Section 2.3, whose components are the fitted values and residuals of a cross-sectional regression on sector indicators.

Both subspaces are invariant under $\boldsymbol{\Sigma}$:

$$\boldsymbol{\Sigma}\, \mathbf{1}_s = k_s \, \mathbf{1}_s, \quad k_s = \sigma_\eta^2 n_s + \sigma_\epsilon^2, \qquad \boldsymbol{\Sigma}\, \mathbf{v} = \sigma_\epsilon^2 \, \mathbf{v} \quad \text{for } \mathbf{v} \in \mathcal{V}, \tag{A.3}$$

where $\mathbf{1}_s$ is the sth column of $\mathbf{B}$ and k_s is the covariance of one contract of sector s with the sum of the sector's contracts. It follows that $\mathbf{M}_W \boldsymbol{\Sigma} \mathbf{M}_V = \mathbf{0}$: no sector-wide vector covaries with a sector-adjusted vector under $\boldsymbol{\Sigma}$.

A.2 Agents

All agents have CARA utility over terminal wealth, so with normal payoffs each maximizes the mean of wealth less half its variance divided by the agent's risk tolerance.

Commercial hedgers. The hedgers of commodity i carry a net physical exposure e_i to its price at $t = 1$. Producers and inventory holders are exposed to a fall in the price and contribute positively to e_i; processors and users with forward commitments are exposed

to a rise and contribute negatively. The sign of e_i is therefore not restricted. Hedgers are represented by one agent with risk tolerance τ_H who chooses futures positions $\mathbf{q}^H$ to maximize the mean-variance objective of $\mathbf{e}^\top \mathbf{P} + \mathbf{q}^{H\top}(\mathbf{P} - \mathbf{F})$.

Unconstrained speculators can hold any futures position $\mathbf{x}^U \in \mathbb{R}^N$ and have risk tolerance τ_U.

Constrained speculators can hold only positions that are neutral within every sector, $\mathbf{x}^C \in \mathcal{V}$. Their risk tolerance is $\tau_C \geq 0$. The constraint describes a relative-value mandate that takes long and short positions across commodities of the same sector but no net sector exposure. The baseline has no constrained speculators, $\tau_C = 0$; every result below holds for any $\tau_C \geq 0$.

Futures are in zero net supply, $\mathbf{q}^H + \mathbf{x}^U + \mathbf{x}^C = \mathbf{0}$.

A.3 Equilibrium

Proposition 1 (Risk premia)**.** *In equilibrium* $\boldsymbol{\pi} = \boldsymbol{\pi}^W + \boldsymbol{\pi}^A$*, with*

$$\pi_s^W = \frac{k_s}{\tau_H + \tau_U}\, e_s^W, \qquad \pi_i^A = \frac{\sigma_\epsilon^2}{\tau_H + \tau_U + \tau_C}\, e_i^A, \tag{A.4}$$

where e_s^W *is the mean net hedging exposure in sector* s *and* $e_i^A = e_i - e_{s(i)}^W$ *is commodity* i*'s exposure relative to its sector. The sector-wide premium depends only on* $\mathbf{e}^W$ *and the sector-adjusted premium only on* $\mathbf{e}^A$*.*

Sector-wide exposure is priced more per unit than sector-adjusted exposure, since $k_s > \sigma_\epsilon^2$ and $\tau_H + \tau_U \leq \tau_H + \tau_U + \tau_C$. With $\tau_C = 0$, the same agents price both components, and the difference comes from the covariance structure alone. With $\tau_C > 0$, constrained speculators also price the sector-adjusted premium, since they cannot absorb sector exposure. Because $k_s = \sigma_\eta^2 n_s + \sigma_\epsilon^2$, the sector-wide premium per unit of mean exposure also rises with the number of commodities in the sector. With only five sectors, each of 7 to 12 contracts, we do not test this implication.

Each component takes the sign of the matching component of net hedging exposure. A sector in which hedgers are net short the physical commodity on average, $e_s^W < 0$, carries a negative sector-wide premium, and a commodity whose hedgers are short relative to their sector, $e_i^A < 0$, a negative sector-adjusted premium. The total premium of a commodity is positive whenever hedgers are net long the physical commodity throughout its sector, $e_j > 0$ for every j in $s(i)$; this is the case of normal backwardation. A commodity with positive exposure in a sector whose exposure is strongly negative can nonetheless carry a negative premium.

Proposition 2 (Returns)**.** *Let* $\mathbf{r} = \mathbf{P} - \mathbf{F}$ *be the vector of futures payoffs per contract. For any sector-wide portfolio* $\mathbf{w}^W \in \mathcal{W}$ *and any sector-adjusted portfolio* $\mathbf{w}^A \in \mathcal{V}$*, with weights fixed at* $t = 0$ *and held to* $t = 1$*,* $\mathrm{Cov}_0(\mathbf{w}^{W\top}\mathbf{r}, \mathbf{w}^{A\top}\mathbf{r}) = 0$.

Proposition 2 follows from the covariance structure (A.2) alone and does not use the equilibrium. It applies to the linear-weight portfolios of Section 3. Their sector-wide weights are the sector means of the ranked signal and lie in $\mathcal{W}$; their sector-adjusted weights are the deviations from them and lie in $\mathcal{V}$. It does not apply to the tercile portfolios, whose sector-adjusted weights need not sum to zero within each sector. The covariance is zero for given weights. Over time, the correlation of the two returns also reflects any comovement of the sector-wide and sector-adjusted premia.

Proposition 3 (Positions)**.** *Equilibrium futures positions are*

$$\begin{aligned}
\mathbf{q}^H &= -\frac{\tau_U}{\tau_H + \tau_U}\,\mathbf{e}^W - \frac{\tau_U + \tau_C}{\tau_H + \tau_U + \tau_C}\,\mathbf{e}^A, \\
\mathbf{x}^U &= \frac{\tau_U}{\tau_H + \tau_U}\,\mathbf{e}^W + \frac{\tau_U}{\tau_H + \tau_U + \tau_C}\,\mathbf{e}^A, \\
\mathbf{x}^C &= \frac{\tau_C}{\tau_H + \tau_U + \tau_C}\,\mathbf{e}^A.
\end{aligned} \tag{A.5}$$

Unconstrained speculators take the other side of hedgers' net positions at both levels. Constrained speculators take the other side of the sector-adjusted component only, and their sector-wide and aggregate positions are zero. With $\tau_C = 0$, $\mathbf{x}^U = -\mathbf{q}^H$, so speculators'

and hedgers' positions have the same sector-wide share of cross-sectional variance. With $\tau_C > 0$, the sector-wide share of the cross-sectional variance of unconstrained speculators' positions exceeds that of hedgers' positions. The two hold equal and opposite sector-wide positions, but unconstrained speculators take only part of hedgers' sector-adjusted position. Hedgers' futures position in the proposition is net of long and short hedging within each commodity, which is what the Producer/Merchant/Processor/User category of the CFTC reports measures.

A.4 Characteristics

Let Ω_i denote the economic conditions that determine commodity i's net hedging exposure, $e_i = h(\Omega_i)$, and let a characteristic be any function of the same conditions observed with noise, $z_i = g(\Omega_i) + \xi_i$, with ξ_i independent of Ω. The characteristic is decomposed across the N commodities as $z_i = z^W_{s(i)} + z^A_i$, with z^W_s the mean of z over sector s. By construction z^A sums to zero within each sector, so the two components are uncorrelated across commodities for any g. The decomposition places no restriction on the functional form of g.

Corollary 1 (Characteristics and premia)**.** *Across commodities, with equal weights,* $\mathrm{Cov}(z^W, \pi^W) = \mathrm{Cov}(z^W, k \circ e^W)/(\tau_H + \tau_U)$ *and* $\mathrm{Cov}(z^A, \pi^A) = \sigma^2_\epsilon \,\mathrm{Cov}(z^A, e^A)/(\tau_H + \tau_U + \tau_C)$*, where* $k \circ e^W$ *is the vector with element* $k_{s(i)} e^W_{s(i)}$*. The cross terms* $\mathrm{Cov}(z^W, \pi^A)$ *and* $\mathrm{Cov}(z^A, \pi^W)$ *are zero.*

A characteristic can therefore be informative about one component of the premium and not the other. Its sector-wide component covaries with sector-wide premia when it covaries with hedging exposure across sectors, and its sector-adjusted component covaries with sector-adjusted premia when it covaries with hedging exposure within sectors.

A.5 Business-Cycle Implications

Consider a recession and an expansion that differ only in speculators' risk-bearing capacity (Acharya et al., 2013; He and Krishnamurthy, 2013). Let the risk tolerance of unconstrained

speculators fall by the proportion x_U in the recession and that of constrained speculators by x_C, holding τ_H and $\mathbf{e}$ fixed. To first order, by Proposition 1, the magnitudes of the premia rise by

$$d\ln|\pi_s^W| = \frac{x_U\tau_U}{\tau_H+\tau_U}, \qquad d\ln|\pi_i^A| = \frac{x_U\tau_U + x_C\tau_C}{\tau_H+\tau_U+\tau_C}, \tag{A.6}$$

and, when $\tau_C > 0$, the sector-wide premium rises proportionally more than the sector-adjusted premium if and only if

$$\frac{x_C}{x_U} < \frac{\tau_U}{\tau_H+\tau_U}. \tag{A.7}$$

With a single speculator type ($\tau_C = 0$) the two expressions in (A.6) coincide, and a change in capacity moves both premia in the same proportion. A sector-wide premium that rises proportionally more in recessions can arise from capacity only when a constrained clientele is present ($\tau_C > 0$) and its capacity is proportionally less cyclical than that of the unconstrained clientele, as stated in (A.7). If both fall in the same proportion, the sector-adjusted premium rises proportionally more.

Which premium rises more also depends on the form of the constraint. If the constrained clientele could also hold sector exposure, it would absorb part of the sector-wide exposure as well. Both premia would then be priced by all three groups, and the two expressions in (A.6) would coincide.

With $\mathbf{e}$ fixed, both premia rise in magnitude when capacity falls. Hedgers hedge less, since by Proposition 3 their positions are proportional to $\tau_U/(\tau_H+\tau_U)$ and $(\tau_U+\tau_C)/(\tau_H+\tau_U+\tau_C)$. In the baseline, the two components can therefore move differently over the cycle only through hedging exposure, and larger hedging positions in recessions also require exposure to rise. Each π_i^A is proportional to e_i^A, so the sector-adjusted premia fall in recessions if the within-sector dispersion of $\mathbf{e}^A$ compresses enough to offset the fall in capacity. A rise in sector-wide exposure raises the sector-wide premia. The model takes this compression as an assumption about hedging exposure, (A5) in Section A.6.

A.6 Assumptions and Results

Assumed. (A1) The factor structure (A.1): one shock per sector with equal loadings within the sector, and commodity shocks that are independent and have equal variance within the sector. (A2) CARA utility and normal payoffs over a single period, with hedgers represented by one agent. (A3) Commercial hedgers carry a net physical exposure of either sign. (A4) Optionally, a second clientele can hold only sector-neutral positions ($\tau_C > 0$); the baseline has $\tau_C = 0$. (A5) The within-sector dispersion of hedging exposure compresses in recessions, which the sector-adjusted premia need in order to fall. With $\tau_C > 0$, a sector-wide premium that rises proportionally more in recessions can also arise from capacity, if the constrained clientele's capacity is proportionally less cyclical than the unconstrained clientele's (Equation (A.7)).

Derived. The separation of premia into a sector-wide component that depends only on sector-wide hedging exposure and a sector-adjusted component that depends only on sector-adjusted exposure, with sector-wide exposure priced more per unit (Proposition 1); the sign of each premium component from the sign of the matching component of net hedging exposure; the pattern of positions across trader types, including equal sector-wide shares of position variance for hedgers and speculators when $\tau_C = 0$ (Proposition 3); and the response of the premia to risk-bearing capacity (A.6).

Following from the assumed covariance structure. The zero conditional covariance between sector-wide and sector-adjusted portfolio returns (Proposition 2) and the zero cross terms in Corollary 1.

Assumption (A1) can be relaxed in two directions without affecting the separation of premia or Proposition 2, since in both cases $\mathbf{\Sigma}$ still maps $\mathcal{W}$ and $\mathcal{V}$ into themselves. Sector shocks may be correlated across sectors, in which case each sector-wide premium depends on the sector-wide exposures of all sectors; and commodity-shock variances may differ across sectors, in which case σ_ϵ^2 in (A.4) is replaced by the variance of the commodity's sector.

The results fail when loadings or commodity-shock variances differ within a sector, because $\boldsymbol{\Sigma}$ then no longer maps $\mathcal{W}$ and $\mathcal{V}$ into themselves, and each premium component depends on both components of exposure. The data depart from (A1) in this respect: within-sector correlations are heterogeneous (Section IA.A of the Internet Appendix), for example between natural gas and the petroleum contracts.

A.7 Proofs

Demands. With mean-variance objectives, hedgers choose $\mathbf{q}^H = \tau_H \boldsymbol{\Sigma}^{-1}\boldsymbol{\pi} - \mathbf{e}$ and unconstrained speculators $\mathbf{x}^U = \tau_U \boldsymbol{\Sigma}^{-1}\boldsymbol{\pi}$. Constrained speculators maximize $\mathbf{x}^\top \boldsymbol{\pi} - \mathbf{x}^\top \boldsymbol{\Sigma}\mathbf{x}/(2\tau_C)$ subject to $\mathbf{B}^\top \mathbf{x} = \mathbf{0}$. The first-order condition is $\mathbf{x} = \tau_C \boldsymbol{\Sigma}^{-1}(\boldsymbol{\pi} - \mathbf{B}\boldsymbol{\mu})$ for multipliers $\boldsymbol{\mu}$. By (A.3), $\mathbf{B}^\top \boldsymbol{\Sigma}^{-1} = \mathbf{D}^{-1}\mathbf{B}^\top$ with $\mathbf{D} = \operatorname{diag}(k_s)$, so the constraint gives $\boldsymbol{\mu} = (\mathbf{B}^\top \mathbf{B})^{-1}\mathbf{B}^\top \boldsymbol{\pi}$ and $\boldsymbol{\pi} - \mathbf{B}\boldsymbol{\mu} = \mathbf{M}_V \boldsymbol{\pi}$. Since $\boldsymbol{\Sigma}^{-1}$ acts on $\mathcal{V}$ as multiplication by σ_ϵ^{-2}, $\mathbf{x}^C = \tau_C\, \sigma_\epsilon^{-2}\, \mathbf{M}_V \boldsymbol{\pi}$.

Proposition 1. Market clearing gives $(\tau_H + \tau_U)\boldsymbol{\Sigma}^{-1}\boldsymbol{\pi} + \tau_C \sigma_\epsilon^{-2} \mathbf{M}_V \boldsymbol{\pi} = \mathbf{e}$. Because $\boldsymbol{\Sigma}^{-1}$ commutes with $\mathbf{M}_W$ and $\mathbf{M}_V$, projecting on $\mathcal{W}$ gives $(\tau_H + \tau_U)\boldsymbol{\Sigma}^{-1}\boldsymbol{\pi}^W = \mathbf{e}^W$, hence $\boldsymbol{\pi}^W = \boldsymbol{\Sigma}\mathbf{e}^W/(\tau_H + \tau_U)$, and (A.3) gives the first expression in (A.4). Projecting on $\mathcal{V}$ gives $(\tau_H + \tau_U + \tau_C)\sigma_\epsilon^{-2}\boldsymbol{\pi}^A = \mathbf{e}^A$, the second expression.

Proposition 2. $\operatorname{Cov}_0(\mathbf{w}^{W\top}\mathbf{r}, \mathbf{w}^{A\top}\mathbf{r}) = \mathbf{w}^{W\top}\boldsymbol{\Sigma}\mathbf{w}^A = \sigma_\epsilon^2\, \mathbf{w}^{W\top}\mathbf{w}^A = 0$, using (A.3) and the orthogonality of $\mathcal{W}$ and $\mathcal{V}$.

Proposition 3. From Proposition 1, $\boldsymbol{\Sigma}^{-1}\boldsymbol{\pi} = \mathbf{e}^W/(\tau_H + \tau_U) + \mathbf{e}^A/(\tau_H + \tau_U + \tau_C)$ and $\sigma_\epsilon^{-2}\mathbf{M}_V\boldsymbol{\pi} = \mathbf{e}^A/(\tau_H + \tau_U + \tau_C)$. Substituting into the demands gives (A.5).

Corollary 1. The two covariances follow from (A.4). $\operatorname{Cov}(z^W, \pi^A) = 0$ because π^A sums to zero within each sector while z^W is constant within it, and $\operatorname{Cov}(z^A, \pi^W) = 0$ symmetrically.

Equation (A.6). $|\pi_s^W| \propto (\tau_H + \tau_U)^{-1}$ and $|\pi_i^A| \propto (\tau_H + \tau_U + \tau_C)^{-1}$; differentiating the logarithms with $d\tau_U = -x_U \tau_U$ and $d\tau_C = -x_C \tau_C$ gives (A.6), and, for $\tau_C > 0$, comparing the two expressions gives (A.7).